GEONOVA

The project explained in simple words, for the whole team

In one sentence: We use free satellite photos to measure how greenery, water and buildings change in three Indian cities (Ahmedabad, Surat, Pune) over 2021 to 2025, and we also test whether two different computer tools (Pandas and Spark) give the same answers and how fast each one is.

1. What problem are we working on?

Cities change. Trees disappear, new buildings appear, lakes grow or shrink. Satellites photograph the same place every few days, so in theory we can watch these changes. In practice there are two difficulties:

- **The environment question:** How do we turn satellite pictures into simple numbers that show change, and can we trust those numbers when clouds, seasons or missing photos get in the way?
- **The computer question:** Satellite data becomes very large. Which tool processes it best on a normal computer, and do two tools give exactly the same results?

GEONOVA studies both questions together in one careful, repeatable plan.

2. What data do we use?

We use **Sentinel-2**, a pair of European satellites run by the Copernicus programme. The images are free from the Copernicus Data Space website. Each photo is taken in many colours of light, including some our eyes cannot see (near-infrared and short-wave infrared). The satellites revisit the same place about every 5 days. We use the "Level-2A" version, where the atmosphere's effect has already been removed, and it also comes with a map that marks clouds and shadows.

3. The three "scores" we calculate

From the colours in each pixel we calculate three simple scores. Each one is a number between -1 and +1.

Score	What it tries to show	Simple idea	Warning
NDVI	Greenness of plants	Healthy plants reflect a lot of near-infrared light and absorb red light	Not a direct measure of trees or crops. Dry season can look like loss
NDWI	Open water	Water reflects green light but absorbs near-infrared	Buildings and shadows can be confused with water
NDBI	Built-up surfaces	Concrete and roads reflect more short-wave infrared than near-infrared	Bare soil and dry land can look like buildings

Important: These scores are *hints*, not proof. If NDVI goes down, it may mean trees were cut, but it may also mean it was a dry season. We never claim "deforestation" or "urbanisation" from these numbers alone.

4. The computer side: Pandas versus Spark

Once the scores are calculated we put them in a big table (millions of rows). Then we process that table with two tools:

- **Pandas** is like **one very good cook** working alone. Simple, fast for small jobs, but limited by one computer's memory.
- **Spark (PySpark)** is like a **team of cooks** sharing the work. It can handle bigger jobs, but organising the team takes time, so it is not always faster.

We run Spark on a **single Windows 11 laptop or PC**. That uses several processor cores, but it is **not** the same as a real cluster of many computers, and we will say so clearly. We do not assume Spark will win. We measure it.

We compare: how long each takes, how much memory it uses, and whether the final numbers are the same.

5. How the project works, step by step

Step	What happens
1	Download Sentinel-2 images for the three cities and check them (dates, version, quality)
2	Clean the images: remove clouds and shadows, line up the colour bands, cut out each city
3	Combine several photos from the same season so the picture is as clear as possible
4	Calculate NDVI, NDWI and NDBI
5	Save everything as one shared table (a Parquet file)
6	Process that same table with Pandas and with Spark, doing exactly the same calculations
7	Compare speed, memory and whether answers match
8	Study the changes across cities and years, and test how clouds and missing photos affect them
9	Store results in a database and show them in a web dashboard (the GEONOVA dashboard)

Tools: **Rasterio and NumPy** handle the satellite pictures, **Pandas and PySpark** handle the big table, **Parquet** is the file format, **PostgreSQL/PostGIS** stores results and map shapes, **FastAPI and React** build the dashboard.

6. The four questions we want to answer

Question	In plain words
RQ1	How do greenery, water and built-up scores differ between the three cities and across the years?

Question	In plain words
RQ2	As the data gets bigger, how does the time and memory of Pandas and Spark change?
RQ3	Do Pandas and Spark give the same numbers when given the same data?
RQ4	How much do clouds, missing photos and seasons change our results?

7. The four experiments

Experiment	What we do	Answers
A	Calculate scores for each city and year, make maps, graphs and tables	RQ1
B	Run Pandas and Spark on 100 thousand, 1 million and 5 million records and time them	RQ2
C	Change the cloud rules, remove some photos, shift the season, and see how results move	RQ4
D	Compare Pandas and Spark outputs number by number	RQ3

8. Our honesty rules

- No made-up numbers, scenes or results. Anything not measured yet is marked **"pending"**.
- Fake (synthetic) data is allowed only for speed tests and is always labelled. All environmental findings use real satellite data.
- We do not say Spark is faster until we have measured it.
- We do not claim a real-world event (like deforestation) without checking against better evidence.
- Every reference must be a real paper that we have checked.

9. Where are we now?

- **Done:** a full paper draft with the plan, the formulas, the diagrams and a literature review. It contains **no results** because experiments have not been run.
- **Next:** download the satellite data, build the Pandas and Spark pipelines, run experiments A to D, then fill the result tables and finish the paper.
- **Open decision for the team:** use a 20 m pixel grid (our default) or 10 m. See the comment in the paper.

10. Quick glossary

Word	Meaning
Pixel	One tiny square of a satellite image
Band	One colour of light recorded by the satellite

Word	Meaning
Cloud mask	A map that marks which pixels are covered by cloud or shadow so we can ignore them
Composite	One clean picture made by combining several photos
Parquet	A compact file format for big tables
Benchmark	A fair speed and memory test
Valid pixels	Pixels we trust (not cloudy, not missing)

The full technical paper is in the GEONOVA document in Claude. Ask there if any step is unclear.